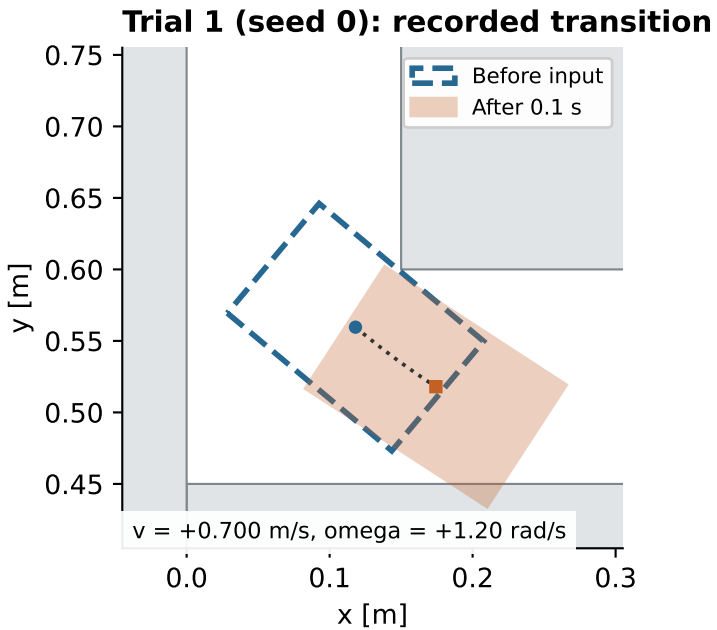
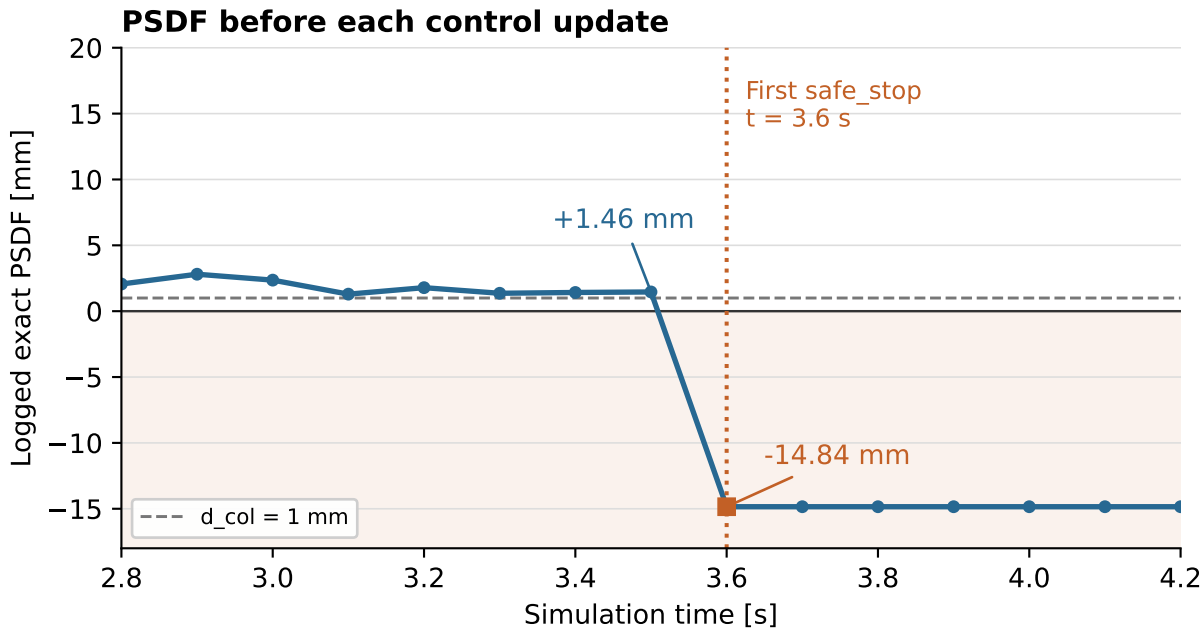
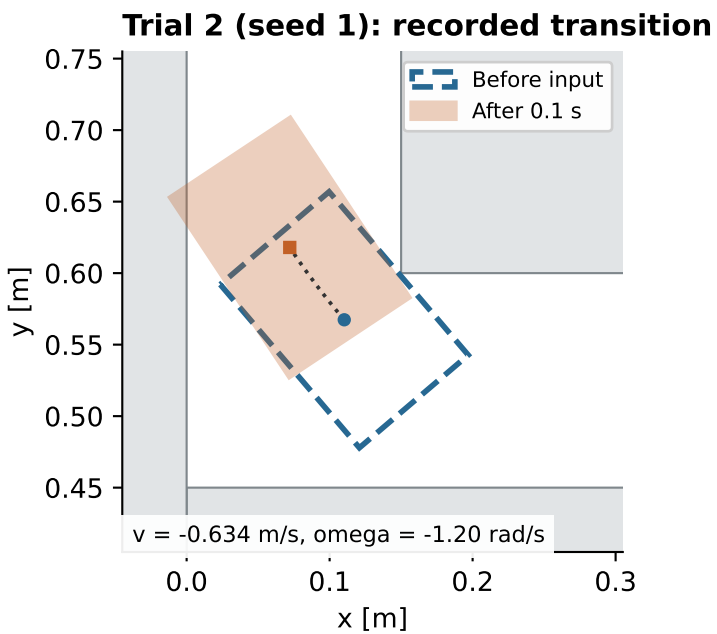
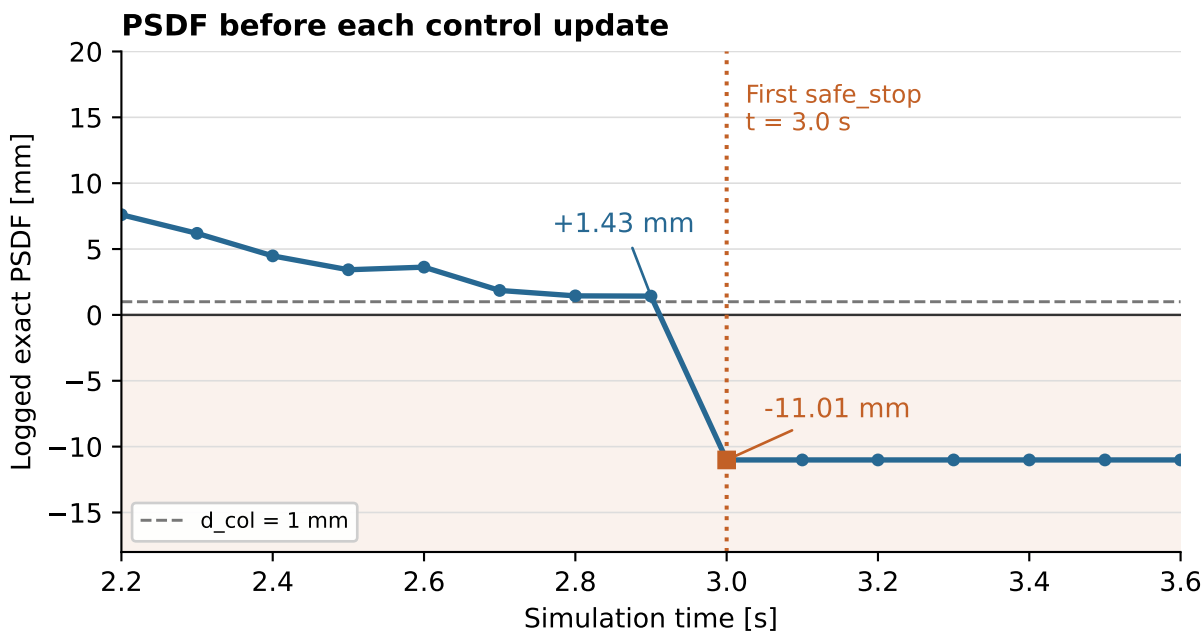
A successful solver step precedes both geometric failures

Saved states and inputs; rectangle 0.15 x 0.10 m; dt = 0.1 s; no controller rerun









Geometry: dashed outline = before input; filled outline = after input. Negative PSDF is confirmed by strict polygon overlap.
Right panels show pre-control samples: a newly unsafe state is visible at the next cycle. Dotted path replays the saved input only.